EE3621: Digital Signal Processing

Lecture 25: DSP in Communications — Modulation, Detection & Matched Filter
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Digital Modulation Review (BPSK, QPSK, QAM).
- **05:00 – 12:00 (7 mins):** Pulse Shaping & Inter-Symbol Interference (ISI); Raised Cosine and Square-Root Raised Cosine filters.
- **12:00 – 22:00 (10 mins):** Matched Filter & Correlation Receiver; SNR maximization proof.
- **22:00 – 28:00 (6 mins):** Equalization techniques (Zero-Forcing, MMSE, DFE).
- **28:00 – 35:00 (7 mins):** OFDM Fundamentals, Cyclic Prefix, FFT/IFFT architecture, and Standards.
- **35:00 – 40:00 (5 mins):** Checkpoints & Worked Examples.

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2 Digital Modulation Review

In modern communication systems, digital information is transmitted by modulating the amplitude and phase of a carrier.

2.1 Complex Baseband Representation

Any passband signal can be mathematically represented in terms of its complex baseband equivalent. Let the passband signal be $x(t)$:

$$x(t) = \mathrm{Re}\{s(t)e^{j2\pi f_c t}\} \tag{1}$$

In discrete time, a modulated symbol is sampled as $s_i[n]$:

$$s_i[n] = I_i[n] + jQ_i[n] \tag{2}$$

where $I_i[n]$ is the In-phase component and $Q_i[n]$ is the Quadrature component.

2.2 Constellations

- **BPSK:** Symbols from $\{+A, -A\}$ (1 bit/symbol).
- **QPSK:** Symbols from $\{\pm A \pm jA\}$ (2 bits/symbol).
- **QAM:** Varies amplitude and phase, $\log_2(M)$ bits/symbol.

3 Pulse Shaping and ISI

3.1 The ISI Problem

The received signal is:

$$y(nT) = s[n]p(0) + \sum_{k \neq n} s[k]p((n-k)T) \tag{3}$$

The second term is the Inter-Symbol Interference (ISI).

3.2 Nyquist Criterion for Zero-ISI

To achieve zero ISI, the time-domain pulse must satisfy:

$$p(nT) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases} \tag{4}$$

3.3 Raised Cosine Filter

The absolute bandwidth required is given by:

$$B = \frac{1+\alpha}{2T} \tag{5}$$

where $\alpha \in [0, 1]$ is the roll-off factor.

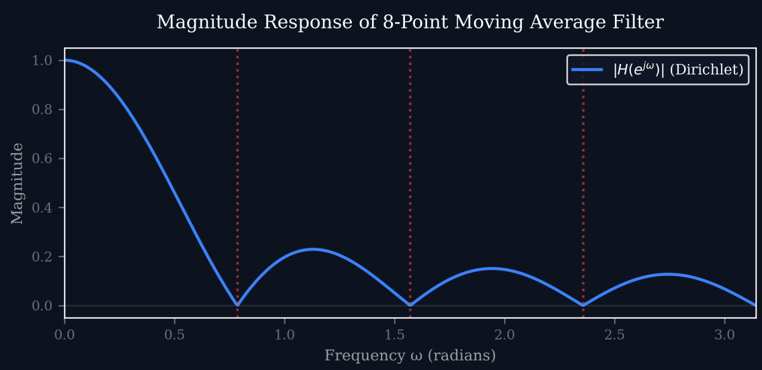

Figure 1: Generic Filter Magnitude Response.

4 The Matched Filter

The Matched Filter maximizes the Signal-to-Noise Ratio (SNR) at the sampling instant.

$$h(t) = s^*(T_0 - t) \tag{6}$$

Discrete form:

$$h[n] = s^*[L-n] \tag{7}$$

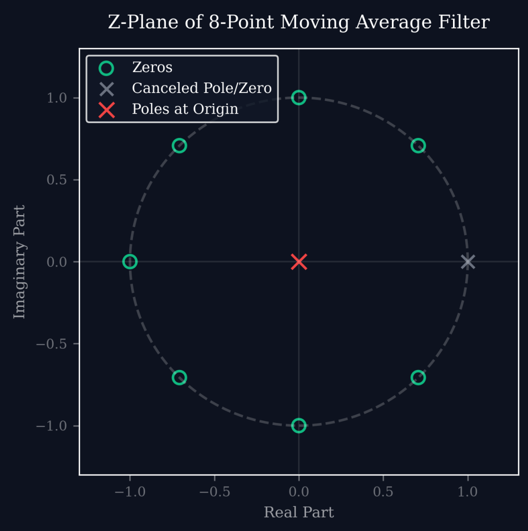

Figure 2: Z-Plane Filter Representation.

4.1 Formal Proof of SNR Maximization

The SNR at the sampling instant is:

$$\mathrm{SNR} = \frac{\left| \int_{-\infty}^{\infty} H(f) S(f) e^{j2\pi f T_0} df \right|^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df} \tag{8}$$

Applying the Cauchy-Schwarz inequality:

$$\left| \int H(f) S(f) e^{j2\pi f T_0} df \right|^2 \leq \left(\int |H(f)|^2 df \right) \left(\int |S(f)|^2 df \right) \tag{9}$$

Equality is achieved when $H(f) = kS^*(f)e^{-j2\pi f T_0}$. The maximum SNR is:

$$\mathrm{SNR}_{\mathrm{max}} = \frac{2E_s}{N_0} \tag{10}$$

5 Channel Equalization

- **Zero-Forcing:** $H_{eq}(f) = 1/H_{channel}(f)$ (Enhances noise).
- **MMSE:** $H_{MMSE}(f) = \frac{H^*(f)}{|H(f)|^2 + N_0/E_s}$.
- **DFE:** Uses past decisions to subtract ISI cleanly.

6 OFDM Fundamentals

Orthogonal Frequency Division Multiplexing divides the high-rate stream into many parallel low-rate subcarriers.

- **Cyclic Prefix (CP):** Converts linear convolution to circular convolution, eliminating ISI.
- **IFFT/FFT:** Used as efficient modulator and demodulator.

7 Summary of Key Formulas

Concept	Formula
Complex Baseband	$s_i[n] = I_i[n] + jQ_i[n]$
Raised Cosine Bandwidth	$B = \frac{1+\alpha}{2T}$
Matched Filter	$h[n] = s^*[L-n]$
Matched Filter SNR	$\text{SNR} = 2E_s/N_0$
MMSE Equalizer	$H_{MMSE}(f) = \frac{H^*(f)}{\lvert H(f)\rvert^2+N_0/E_s}$

8 Checkpoints

Detailed step-by-step solutions for Checkpoints are available in the full markdown notes.